

EEE 359

# Engineering Project Management

# Lecture 10

# Risk Management

# Definition

An uncertain event or set of events that, should it occur, will have an effect on the achievement of objectives. A risk is measured by a combination of the **probability** of a perceived threat or opportunity occurring, and the magnitude of its impact on objectives.

Risks can have either a negative or positive impact on objectives if they occur.

- **Threat:** For uncertain events that would have a negative impact on objectives.
- **Opportunity:** For uncertain events that would have a positive impact on objectives.

These can impact the project's objectives of delivering an agreed scope and benefits to an agreed time, cost and quality.

# Overview

1. Risk management is an ongoing process
2. Done frequently from **initiating to closing**
3. Risks can be **identified at any time** during the project
4. If uncovered after the initial risk identification process a risk **MUST always be qualitatively analyzed (ranked)**
5. Risk processes must be done in the correct order
6. **Log potential risk response** at any time
7. If serious or urgent, a response strategy is planned

# Sources of Risk

- Customer
- Vendors (*including 2<sup>nd</sup> tier*)
- Stakeholders resisting change
- Geographic, cultural, or language differences
- Lack of project management knowledge
- Not enough effort given to project management
- Regulatory and government agencies
- Senior management constraints
- Basically -- ANYWHERE

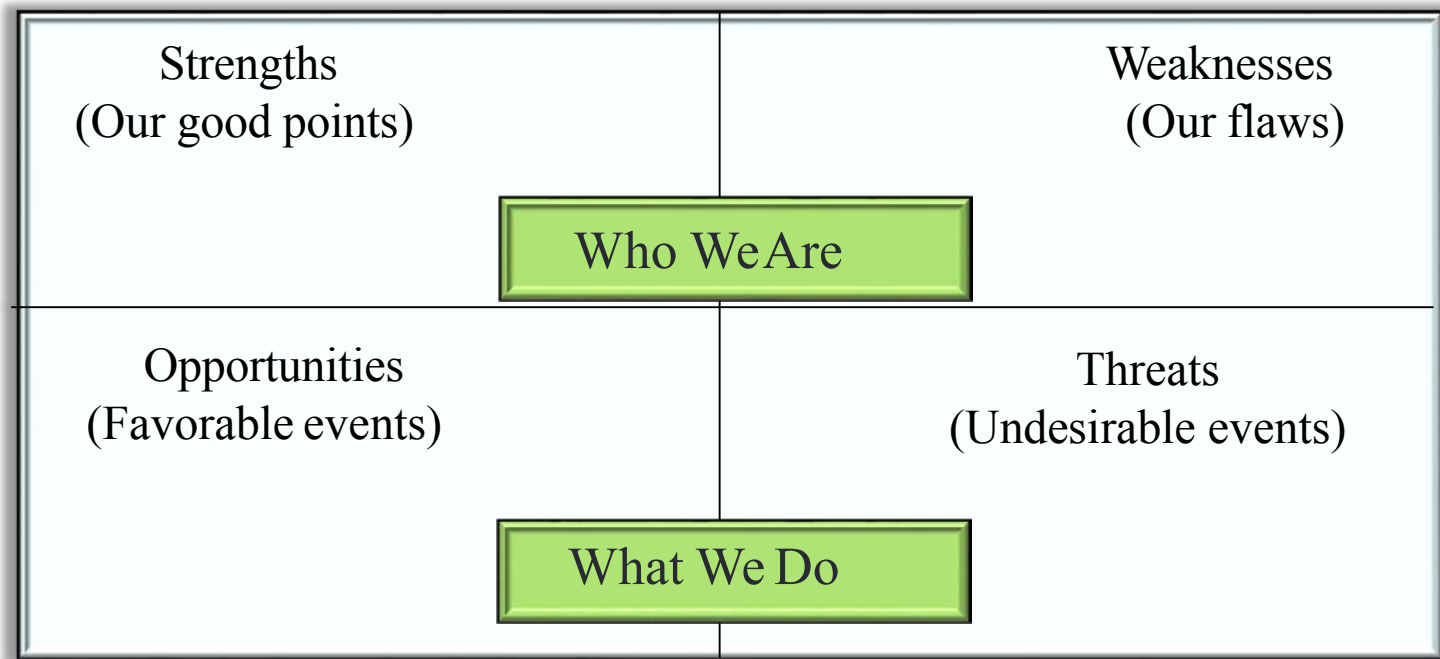


# Risk Management Procedure



# SWOT Analysis

Strengths, Weaknesses, Opportunities, and Threats



# SWOT Analysis

| <div>S</div> <div>STRENGTHS</div>                                                                                                                                                                                                                                                                             | <div>W</div> <div>WEAKNESSES</div>                                                                                                                                                                           | <div>O</div> <div>OPPORTUNITIES</div>                                                                                                                                                                                                   | <div>T</div> <div>THREATS</div>                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Things your company does well</li><li>• Qualities that separate you from your competitors</li><li>• Internal resources such as skilled, knowledgeable staff</li><li>• Tangible assets such as intellectual property, capital, proprietary technologies etc.</li></ul> | <ul style="list-style-type: none"><li>• Things your company lacks</li><li>• Things your competitors do better than you</li><li>• Resource limitations</li><li>• Unclear unique selling proposition</li></ul> | <ul style="list-style-type: none"><li>• Underserved markets for specific products</li><li>• Few competitors in your area</li><li>• Emerging need for your products or services</li><li>• Press/media coverage of your company</li></ul> | <ul style="list-style-type: none"><li>• Emerging competitors</li><li>• Changing regulatory environment</li><li>• Negative press/media coverage</li><li>• Changing customer attitudes toward your company</li></ul> |

# Identification of Risks

- **PESTEL Analysis** - identifies external risks, helps in planning.
- **Stakeholder Analysis** - assessing stakeholder influence and interest, and evaluating how they impact or are impacted by project risks.
- **Probability and Impact Matrix/Risk Assessment Matrix:**
  - A grid for mapping probability of risk occurrence and its impact on the project objectives if that risk occurs.
  - Risks are prioritized according to their potential impacts for project objectives.

# PESTELAnalysis

## PESTEL Analysis

**P**

### **POLITICAL**

- Fiscal policy
- Government activity
- Conflicts / help
- Taxes

**E**

### **ECONOMICAL**

- GDP
- Employment rate
- Exchange rate
- Inflation
- Income level

**S**

### **SOCIAL**

- Demographic variables
- Cultural factor
- Religion
- Lifestyle
- Education level

**T**

### **TECHNOLOGICAL**

- Technological access
- Infrastructure
- Research
- Technology trends

**E**

### **ENVIRONMENTAL**

- Environmental policies
- Recycling
- Consumption trends
- Production processes
- Natural risks

**L**

### **LEGAL**

- Wages
- Rights
- Job security
- Regulations
- Controlled of Marketing

# Stakeholder Analysis

- Some stakeholders **introduce** risks (e.g., suppliers, regulators)
- Some are **affected** by risks (e.g., users, community)
- Some can help **mitigate** risks (e.g., experts, sponsors)

Steps in stakeholder analysis:

1. **Identify stakeholders:** Direct or Indirect.
2. **Analyze stakeholders:** using power-interest and power-influence grids.
3. **Risk identification:** introduce, affected by or mitigate risks?
4. **Risk response:** communication and engagement strategies for risk mitigation.

# Probability and Impact Matrix/Risk Assessment Matrix

| Likelihood | Impact        |         |          |             |        |
|------------|---------------|---------|----------|-------------|--------|
|            | Negligible    | Minor   | Moderate | Significant | Severe |
|            | Very Likely   | Low Med | Medium   | Med Hi      | High   |
|            | Likely        | Low     | Low Med  | Medium      | Med Hi |
|            | Possible      | Low     | Low Med  | Medium      | Med Hi |
|            | Unlikely      | Low     | Low Med  | Medium      | Med Hi |
|            | Very Unlikely | Low     | Low      | Low Med     | Medium |

| Probability | Impact                                                      |           |                 |           |            |
|-------------|-------------------------------------------------------------|-----------|-----------------|-----------|------------|
|             | Measures the potential severity of the risk's consequences. |           |                 |           |            |
|             | Insignificant (1)                                           | Minor (2) | Significant (3) | Major (4) | Severe (5) |
|             | Almost Certain (5)                                          | 5         | 10              | 15        | 20         |
|             | Likely (4)                                                  | 4         | 8               | 12        | 16         |
|             | Moderate (3)                                                | 3         | 6               | 9         | 12         |
|             | Unlikely (2)                                                | 2         | 4               | 6         | 8          |
|             | Rarely (1)                                                  | 1         | 2               | 3         | 4          |

Risk Levels: Low, Moderate, High, Critical

# Probability and Impact Matrix/Risk Assessment Matrix

## Risk Assessment

| Risk               | Probability | Impact | Risk Level  | Risk Score |
|--------------------|-------------|--------|-------------|------------|
| Battery failure    | Medium      | High   | High Risk   | 10-16      |
| Regulatory delay   | High        | High   | Critical    | 17-25      |
| Minor software bug | Low         | Low    | Low Risk    | 1-4        |
| Weather disruption | Medium      | Medium | Medium Risk | 5-9        |

## Risk Mitigation

| Risk Level | Mitigation Strategy            |
|------------|--------------------------------|
| Low        | Monitor                        |
| Medium     | Plan mitigation                |
| High       | Immediate action               |
| Critical   | Urgent attention + backup plan |

# Perform Qualitative Risk Analysis

The process of prioritizing individual project risks for further analysis or action by assessing their probability of occurrence and impact as well as other characteristics. It focuses efforts on high-priority risks.

Tools: Probability-Impact Matrix, Risk categorization, Expert judgement

**ALWAYS do qualitative analysis.**

# Quantitative Risk Analysis

- The process of numerically analyzing the **combined effect** of identified individual project risks and other source of uncertainty on **overall project objectives** such as cost, time, or performance.
- It **quantifies** overall risk exposure, and it can also provide additional quantitative risk information to support risk response planning.
- It uses numerical data and statistical methods to evaluate the probability and impact of risks on a project.
- Unlike qualitative risk analysis (which ranks risks as "high/medium/low"), quantitative analysis **assigns actual numbers** - probabilities, costs, durations - to risks and their potential outcomes.

# Quantitative Risk Analysis

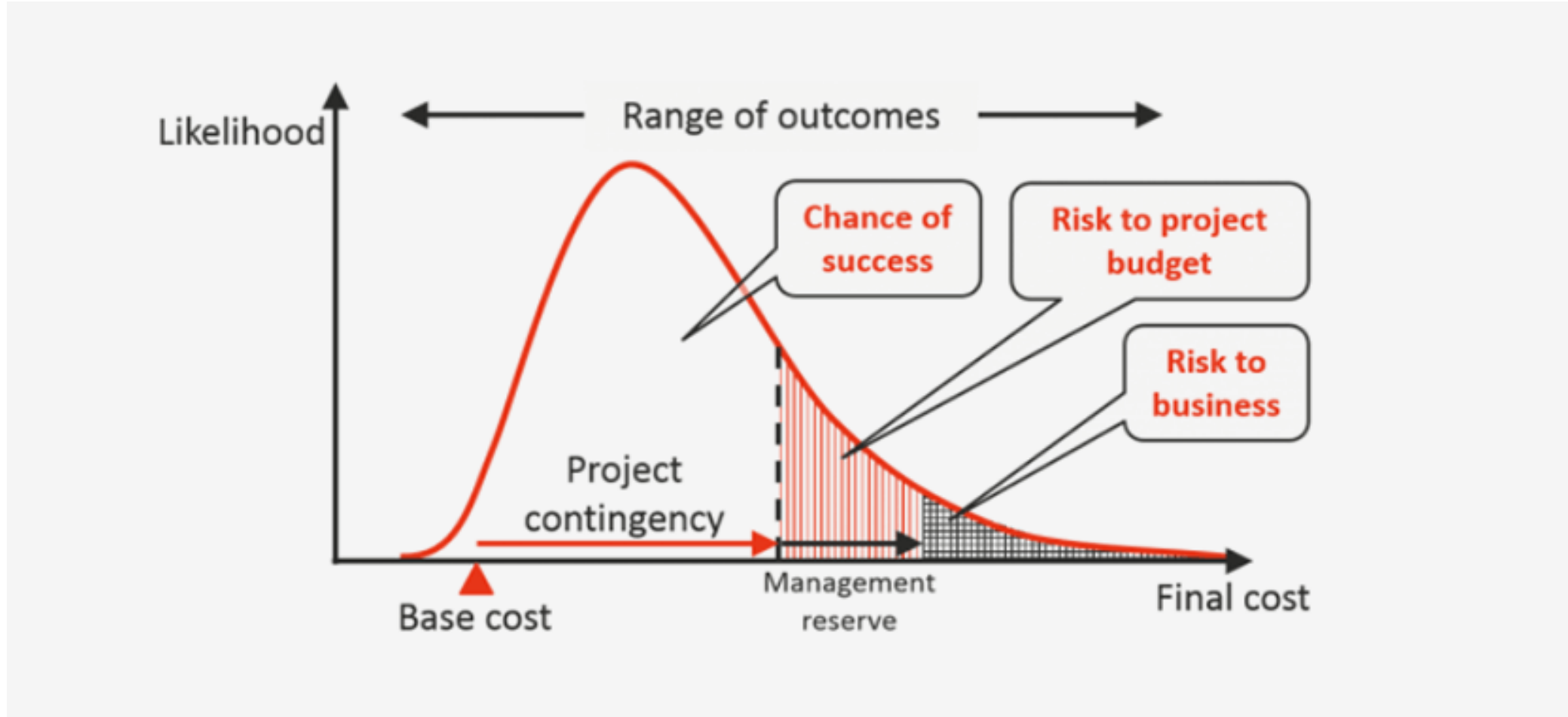
## Purposes:

- Helps justify budget contingencies (schedule and cost reserves) to stakeholders.
- Determine overall project risk (how much risks affect the project).
- Determine the probability of meeting the project objectives, and estimating possible schedule delays or cost overruns.
- Determine which risk events need a response.
- Provides defensible, data-driven risk decision-making.

# Monte Carlo Analysis

- Runs thousands of software simulations using ranges of estimates (optimistic, most likely, pessimistic)
- To produce a probability distribution of possible outcomes.
- Yields the probability of completing the project on a specific scheduled target date and/or a specific target cost.
- E.g., "There's a 80% chance of the project completing within 6 months and under \$10,000."

# Monte Carlo Analysis



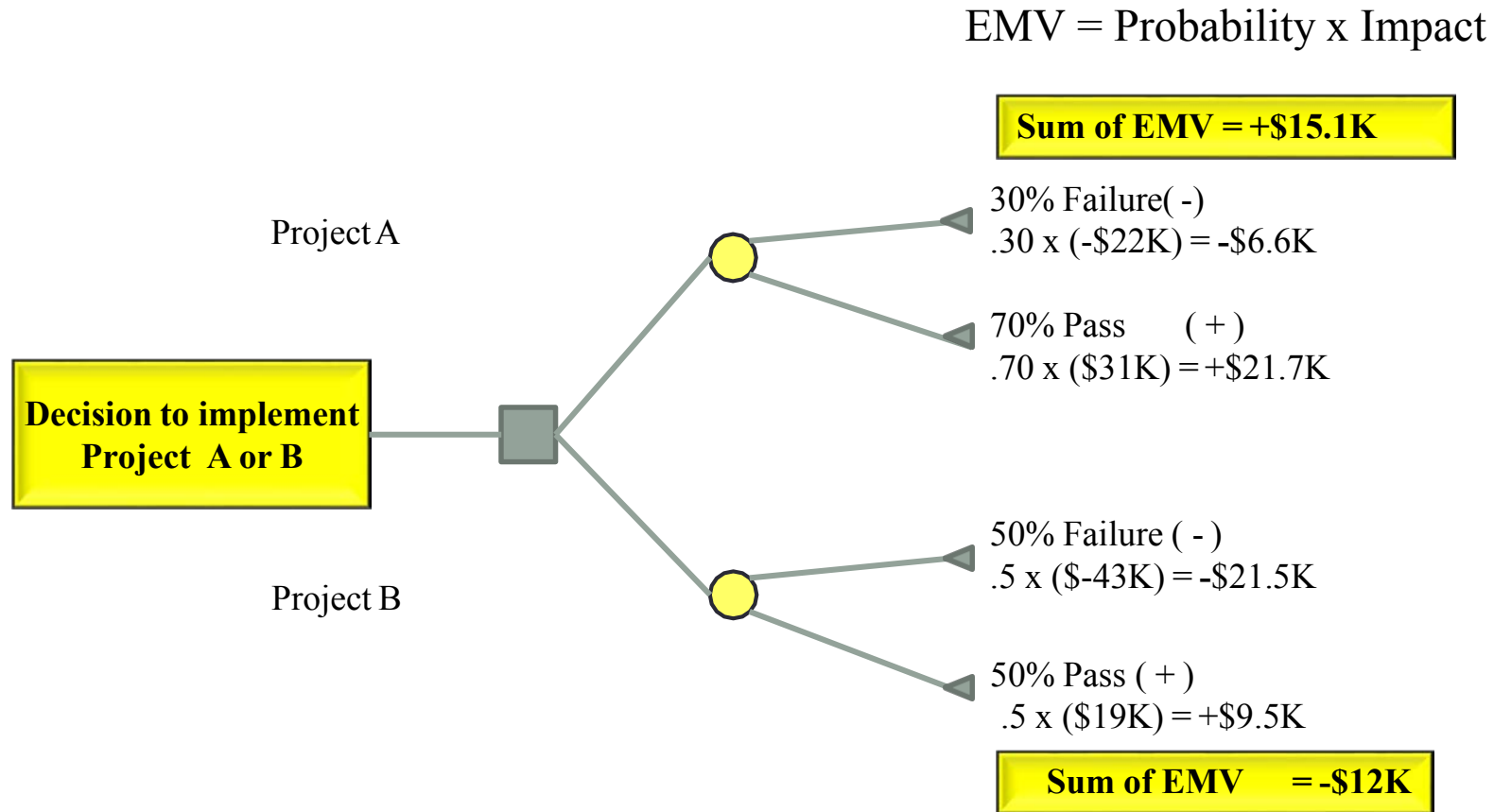
# Expected Monetary Value (EMV)

- Used to estimate the **average expected cost** of a risk.
- **EMV = Probability × Impact**
- A project risk has a 30% chance of causing a \$10,000 cost overrun, what is the EMV?
  - Risk Probability = 0.3
  - Risk Impact = \$10,000
  - **EMV** = 0.3 x 10,000 = \$3,000

# Decision Tree Analysis

- Visual tool to compare different choices under uncertainty.
- Maps out decision points and chance events as a tree.
- With probabilities and values assigned to each branch.
- Uses EMV to select the best option.
- Useful when facing sequential decisions under uncertainty.

# Decision Tree Analysis

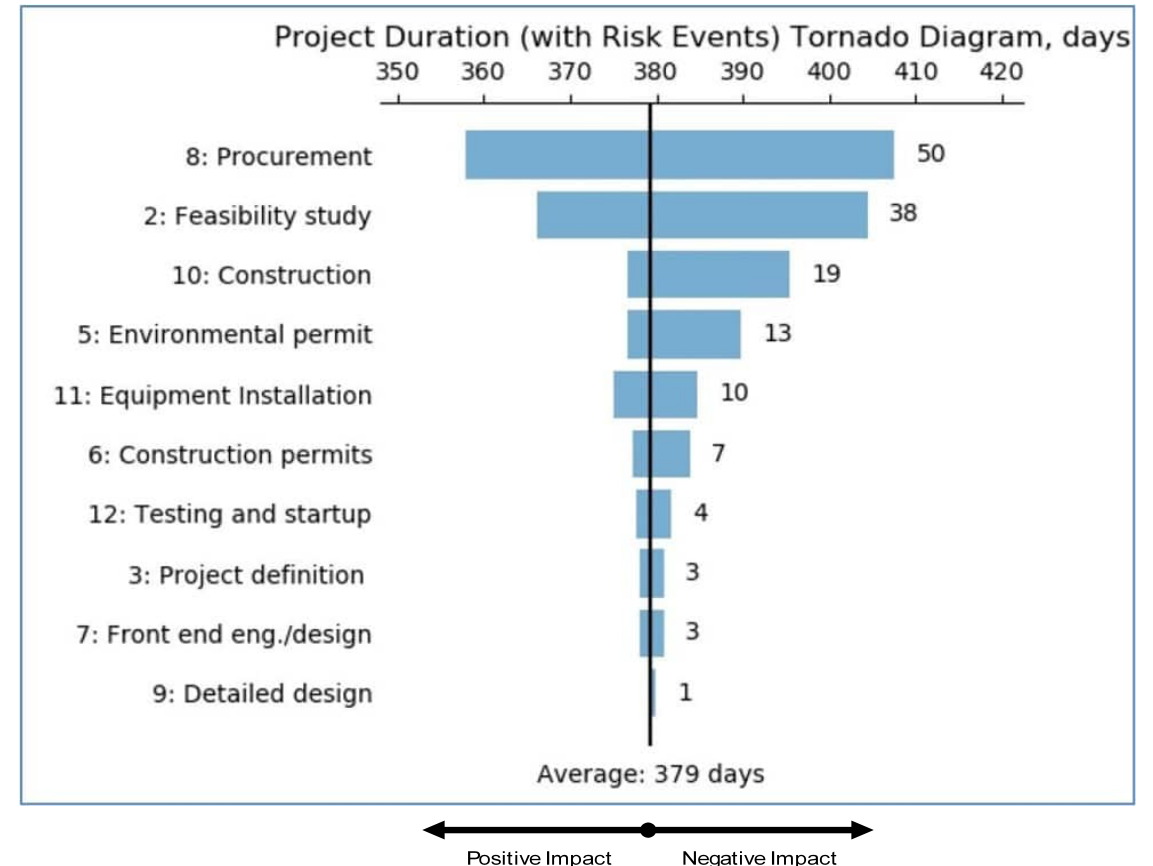
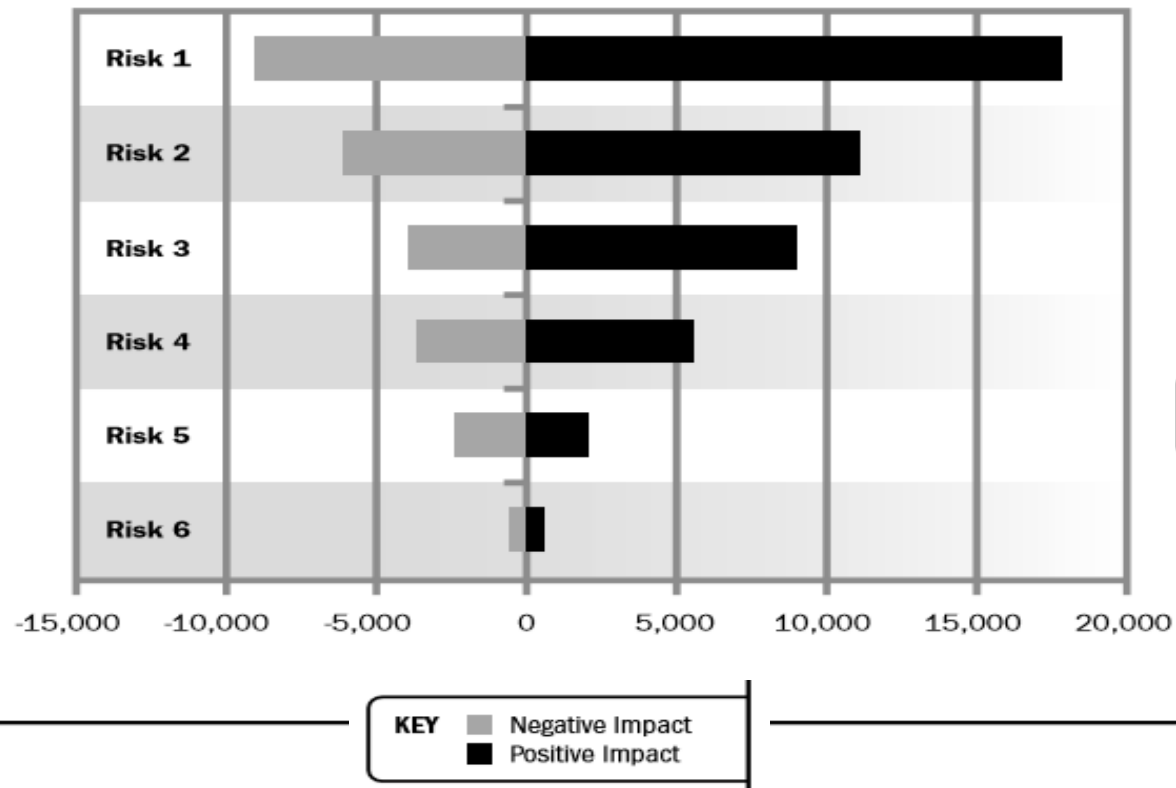


Which Project to choose?

✓ Project A has the higher positive monetary value.

# Sensitivity Analysis

- Determines which risks have the most potential impact on the project outcome.
- A Tornado Diagram is a typical display of sensitivity analysis.



# Risks Monitoring

- The process of monitoring the implementation of agreed-upon risk response plans, tracking identified risks, identifying and analyzing new risks, and evaluating risk process effectiveness throughout the project.
- It enables project decisions to be based on current information about overall project risk exposure and individual project risks.

## Risk Review

Risk reviews may result in identification of new individual project risks, (including secondary risks that arise from agreed-upon risk responses), reassessment of current risks, the closing of risks that are outdated, issues that have arisen as the result of risks that have occurred, and identification of lessons to be learned for implementation in ongoing phases in the current project or in similar projects in the future.

Thanks